

Can Concept Erasure Reproduce Concept Exclusion? A Matched Evaluation of EMBER, RMU, and SNMF

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Study design schematic to draw from final protocol

Figure 1: Planned comparison of the full model, concept-excluded twin, and three erasure methods on identical target, neighboring, and unrelated sentence completions.

Abstract

Plan: Target 150–190 words. State the erasure question; matched full/twin/ EMBER/RMU/SNMF design; three concepts and three evaluation groups; answer NLL and full-vocabulary KL; the most important result and qualification once known. Write last. No numbers until all five models use the same held-out protocol.

1 Introduction

Plan: About 0.7 page. Motivate selective concept erasure and explain why “forgetting” needs two checks: approaching the concept-excluded twin on the target and retaining correct-answer support elsewhere. Define the full model, excluded twin, and post-hoc edit before using those terms repeatedly. Name Ancient Rome, Baseball, and AI. State that the study measures answer-token probability and next-token distributional similarity, not generated-answer accuracy. End with 2–3 concrete contributions once results exist.

2 Background and Related Work

2.1 LMEnt and Concept Exclusion

LMEnt provides a Wikipedia-based pretraining corpus annotated with entity mentions, an index for retrieving chunks associated with those entities, and language models trained on the corpus (Gottesman et al., 2025). The entity index lets us identify training chunks linked to a target concept. By masking the loss on those chunks during an otherwise matched training run, we obtain a *concept-excluded*

twin. This twin provides a reference for the effect of excluding a defined portion of the training data. It may still learn related facts from chunks outside that portion.

2.2 Post-Training Concept Erasure

EMBER edits token embeddings rather than internal transformer layers (Suslik et al., 2026). It factorizes the embedding matrix into sparse features, selects features associated with the target concept, and subtracts their contributions from the embeddings of selected tokens. This makes it possible to change a small set of input representations without updating the rest of the model.

We also consider two approaches that act on multilayer perceptron (MLP) components. Representation Misdirection for Unlearning (RMU) updates selected MLP parameters to redirect representations of forget-set inputs, while a retain objective encourages representations of other inputs to remain close to the original model (Li et al., 2024). Semi-nonnegative matrix factorization (SNMF) identifies interpretable features in MLP activations from groups of co-activated neurons (Shafran et al., 2026). In its erasure adaptation, concept-associated features are mapped to directions that can be removed through MLP weight edits (Suslik et al., 2026).

These methods give us embedding and internal-layer edits to compare with the same concept-excluded reference.

The EMBER study evaluated erasure using multiple-choice and open-ended questions about target concepts, with related-topic questions and general benchmarks used to assess preservation (Suslik et al., 2026). It also examined generation quality and whether target knowledge returned after relearning. These evaluations compared edited models with the original model, which had been trained on the target concept. They included EMBER, RMU, and an MLP weight-erasure adapta-

tion of SNMF-derived features (Li et al., 2024; Shafran et al., 2026). RMU’s original evaluation likewise measured target-question performance and general capabilities relative to the original model. Our study adds a matched concept-excluded twin as a reference and compares edited models with it using correct-answer NLL and full-vocabulary KL on the same sentence completions.

3 Experimental Design

3.1 Shared Base Model and Matched Twins

Plan: About 0.3 page. State architecture, corpus, checkpoint, initialization, batch order, optimizer, duration, and how labels from selected chunks are masked. Explain the shared full model and one twin per concept.

3.2 Target Concepts and Exclusion Scope

Plan: About 0.25 page. Define Ancient Rome, Baseball, and AI blacklist rules; give QID/chunk footprint and any precision/coverage audit. Explicitly connect the target evaluation group to each exclusion scope.

3.3 Erasure Checkpoints and Model Selection

Plan: About 0.35 page. State that EMBER, RMU, and SNMF start from the same full checkpoint. Summarize edited parameter families, training or factorization data, and how settings were selected without using final held-out items. Describe one selected checkpoint per method and concept. Put exact settings and chronology in Appendix A.

3.4 Evaluation Data Separation

Plan: About 0.2 page. Explain separate erasure-training, selection, and final evaluation sentence completions, identical across all five models. Introduce target, neighboring, and unrelated groups. Every item has one fixed correct continuation. Counts and question provenance go in Appendix B.

4 Evaluation Protocol

4.1 Three Groups and Fixed Correct Answers

Plan: About 0.2 page. Define target, neighboring, and unrelated groups and their distinct roles. Each item has a declarative sentence-completion prompt and one fixed correct continuation, for example “The city of Rome is located in” followed by “Italy.” Score only the answer tokens. State explicitly that no candidates are ranked and no text is generated.

4.2 Answer NLL Difference

Plan: About 0.25 page. For each question, average the negative log-probability of each correct-answer token given the prompt and preceding correct-answer tokens. Subtract the reference model’s per-question NLL from the evaluated model’s,

then average question differences. Positive means less probability on the correct answer. A signed mean near zero may conceal cancellation.

4.3 Teacher-Forced Full-Vocabulary KL

Plan: About 0.25 page. At each correct-answer position, compare the full next-token vocabulary distributions under identical teacher-forced prefixes. Compute $KL(\text{reference} \rightarrow \text{evaluated})$, average answer positions within each question, then average questions. KL measures distributional similarity, not whether the correct answer improved, and is not divergence between generated answers.

4.4 Model Comparisons and Uncertainty

Plan: About 0.3 page. Twin minus full establishes the exclusion effect; erasure minus twin measures resemblance; erasure minus full reveals target change and preservation. Report both metrics separately for target, neighboring, and unrelated groups. Define identical prompts, answers, tokenization and context settings. State paired bootstrap unit, paraphrase clustering and intervals; avoid equivalence claims from a near-zero mean.

5 Results

5.1 What Concept Exclusion Changes

Plan: About 0.4 page. For each concept, establish twin-minus-full answer NLL and $KL(\text{full} \rightarrow \text{twin})$ on target questions, then neighboring and unrelated questions. State where the exclusion reference has weak observable effect. Point to complete per-group twin rows in Appendix D.

5.2 How Closely Erasure Matches the Twin on Target Items

Plan: About 0.7 page. Compare EMBER, RMU and SNMF with the twin and full model on target answer NLL and full-vocabulary KL. Show paired intervals and inspect per-question variation; a near-zero signed mean NLL does not establish identical behavior. Discuss substantive differences among Rome, Baseball, and AI only after the new measurements are filled.

5.3 Neighboring and Unrelated Correct-Answer Support

Plan: About 0.6 page. Give neighboring and unrelated answer-NLL and KL comparisons separately for every method. Positive NLL from full indicates weaker prediction of the correct answer; KL indicates a distributional change but has no direction of quality. Discuss neighboring spillover and unrelated preservation relative to both full and twin.

Concept	Chunks	Target Q	Neighbor Q	Unrelated Q
Ancient Rome	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Baseball	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
AI	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>

Table 1: Planned study inventory. Replace placeholders with final masked chunk and held-out completion counts.

Concept	Erasure model	Δ NLL vs full	Δ NLL vs twin	KL(twin $\rightarrow$ erasure)
Rome	EMBER	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	RMU	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	SNMF	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
Baseball	EMBER	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	RMU	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	SNMF	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
AI	EMBER	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	RMU	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>
	SNMF	<i>TBD</i>	<i>TBD</i>	<i>TBD</i>

Table 2: Target-question summary to populate after common-protocol scoring. NLL changes are signed evaluated-minus-reference differences. Show paired intervals in the final version and the twin-minus-full reference effect nearby.

Signed answer-NLL changes by concept, model, and question group

Figure 2: Planned answer-NLL differences from the full model on target, neighboring, and unrelated sentence completions, with paired intervals.

5.4 Where Answer NLL and KL Disagree

Plan: About 0.3 page. Examine cases where the correct-answer probability appears stable but the full next-token distribution changes, or vice versa. Check whether a near-zero mean signed NLL difference hides opposing question-level changes. Use concrete paired examples only if supported by the final data.

6 Analysis and Discussion

6.1 What Counts as Matching Concept Exclusion?

Plan: About 0.35 page. Synthesize target similarity and preservation across answer NLL and KL. Explain why higher target answer NLL than the twin does not establish a better edit, and why low KL alone cannot establish knowledge preservation. Respect the scope of fixed-answer sentence completions.

6.2 Why Concepts and Metrics May Disagree

Plan: About 0.45 page. Investigate supported explanations: blacklist coverage, neighboring-topic overlap, per-question NLL cancellation, and disagreements between correct-answer NLL and full-vocabulary KL. Clearly distinguish observed

evidence from hypotheses. Add a small diagnostic figure only if it clarifies a central result; otherwise put diagnostics in Appendix E.

7 Limitations

Plan: About 0.4 page. State limits from operational blacklist coverage, fixed correct answers rather than generated-answer accuracy, no general passage-prediction measurement, single training seed/checkpoint per arm, finite and potentially dependent questions, and teacher-forced KL. Identify any exploratory selection or unequal evaluation cell honestly when results are filled.

8 Conclusion

Plan: About 0.2 page. Answer which method most closely matches each twin on correct-answer NLL and full-vocabulary KL, and whether neighboring/unrelated answer support is preserved, at the level actually supported. Do not claim generated-answer accuracy or passage-level erasure. One restrained next step.

AI Disclosure and Reflection

Plan: About 0.2 page. Fill with the actual AI tools and models the team used, their roles in coding, editing, or analysis, and a concise reflection on verification. The course guidelines require specific names; do not guess.

References

Daniela Gottesman, Alon Gilae-Dotan, Ido Cohen, Yoav Gur-Arieh, Marius Mosbach, Ori Yoran, and Mor

Geva. 2025. LMEnt: A suite for analyzing knowledge in language models from pretraining data to representations. *Preprint*, arXiv:2509.03405.

Nathaniel Li, Alexander Pan, Anjali Gopal, Summer Yue, Daniel Berrios, Alice Gatti, Justin D. Li, Ann-Kathrin Dombrowski, Shashwat Goel, Long Phan, Gabriel Mukobi, Nathan Helm-Burger, Rassin Lababidi, Lennart Justen, Andrew B. Liu, Michael Chen, Isabelle Barrass, Oliver Zhang, Xiaoyuan Zhu, et al. 2024. The WMDP benchmark: Measuring and reducing malicious use with unlearning. In *Proceedings of the 41st International Conference on Machine Learning*, pages 28525–28550.

Or David Shafran, Atticus Geiger, and Mor Geva. 2026. Constructing interpretable features from compositional neuron groups. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 42326–42348.

Clara Haya Suslik, Or Shafran, and Mor Geva. 2026. Don’t forget your embeddings: Robust knowledge erasure via precise editing of embeddings. *Preprint*, arXiv:2606.03695.

A Corpus Exclusion and Model Configurations

Plan: Table A1: one row per concept and method with checkpoint ID, erasure settings, training/selection data, edited parameters, and seed. Give exact blacklist rules and audit estimates. Do not paste raw run logs.

B Sentence-Completion Evaluation Sets

Plan: Table A2: target, neighboring, and unrelated question counts for each concept; item construction and exclusions; one representative fixed prompt/answer; question provenance and train/selection/test overlap checks.

C Scoring and Statistical Details

Plan: Write exact equations for per-question answer-token NLL, signed evaluated-minus-reference group mean, and full-vocabulary $KL(\text{reference} \rightarrow \text{evaluated})$ under correct-answer teacher forcing. Specify tokenization, prefix, answer-token mask, position/question aggregation, bootstrap resampling and paraphrase clustering. Include one worked item if useful. No option-accuracy or passage-NLL metrics belong in this protocol.

D Complete Results by Concept

D.1 Ancient Rome

Plan: Table A3: all five models, all three question groups, answer NLL and full-vocabulary KL; underlying group means, full/twin contrasts, and paired intervals. Split by metric if needed.

D.2 Baseball

Plan: Table A4 with the identical structure and question-level protocol.

D.3 Artificial Intelligence

Plan: Table A5 with the identical structure and question-level protocol.

E Diagnostic Analyses

Plan: Optional: per-question signed answer-NLL differences showing possible cancellation; examples where NLL and KL disagree; distributional plots and supported case studies. Include only analyses actually performed and needed to interpret main results.

F Reproducibility Notes

Plan: Final scoring script/version, model IDs, seeds, exact question manifest and split chronology, tokenization/context settings, answer scoring masks, and any deviations from the common protocol. Keep this concise.